9. A brand of antiseptic mouthwash contains hydrogen peroxide along with several other chemicals such as water, flavourings and colouring.

(a) 100 cm^3 of mouthwash contains 1·5 g of 35% hydrogen peroxide solution.

A 35% hydrogen peroxide solution contains 35 g of hydrogen peroxide in 100 cm^3 of solution.

(i) Calculate the mass of hydrogen peroxide, in g, present in a 300 cm^3 bottle of the mouthwash. **1**

(ii) Enzymes in saliva act as catalysts in the decomposition of hydrogen peroxide.

Name the family of compounds to which enzymes belong. **1**

(iii) The concentration of hydrogen peroxide can be determined by a titration with a solution of potassium permanganate.

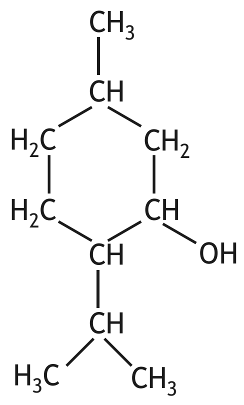
Name the **two** pieces of equipment that would be required to accurately measure the volumes of hydrogen peroxide and potassium permanganate used in the titration. **1**

(b) Volatile, non-water-soluble compounds obtained from plants can be used to provide the minty aroma of the mouthwash.

State the term used to describe the mixture of volatile, non-water-soluble aroma compounds obtained from plants. **1**

9. (continued)

(c) The mouthwash also contains menthol.



menthol

(i) Menthol is based on isoprene units.

State the systematic name for isoprene. **1**

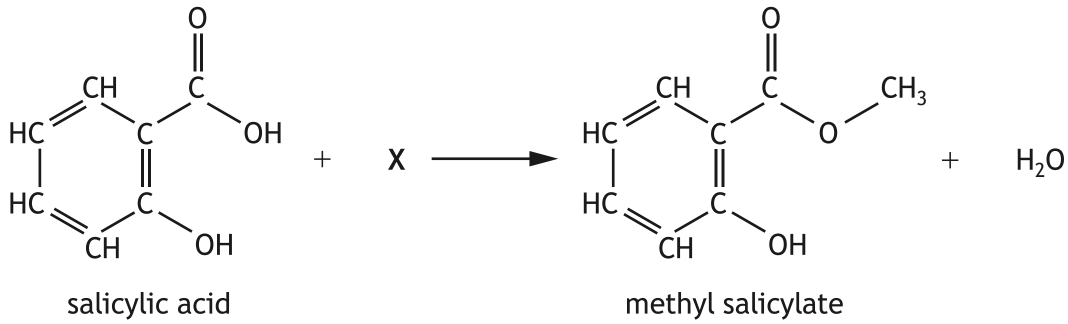
(ii) Menthol can be oxidised to form a mint flavoured compound.

State the type of compound formed when menthol is oxidised. **1**

[Turn over

9. (continued)

(d) Methyl salicylate is an ester found in the mouthwash that can be formed from salicylic acid.



(i) Name reactant **X**. **1**

(ii) A scientist prepared a sample of methyl salicylate using 28·3 g salicylic acid and an excess of reactant **X**.

$C_7H_6O_3$	+	**X**	$\rightarrow$	$C_8H_8O_3$	+	H_2O
salicylic acid				methyl salicylate		
$GFM = 138$ g				$GFM = 152$ g		

The scientist produced 24·7 g of methyl salicylate.

Calculate the percentage yield of methyl salicylate. **2**

9. (d) (continued)

(iii) Methyl salicylate can be toxic to humans at 0·14 g per kg of body mass. It can be obtained from a plant substance called oil of wintergreen. 5·0 cm^3 of oil of wintergreen contains 7·0 g of methyl salicylate.

Calculate the minimum volume, in cm^3, of oil of wintergreen that would provide a toxic dose to a human with body mass of 65 kg. **2**

[Turn over